



CENTRAL POWER RESEARCH INSTITUTE

(Member of STL)

TEST REPORT

Test Report Number : CPRIBLRSCCL25T1654 Date: 10-10-2026

Name and Address of the Customer : A.P. Transformers, SP24 & 95H/1 (SP), Ambattur Indl Estate, Chennai - 600058

Name and Address of the Manufacturer : NA

Particulars of sample tested : 250 kVA, 11000 / 433 V, Three Phase Distribution Transformer

Type : ONAN outdoor type distribution transformer

Description of test sample : Refer Sheet 2 of 11

Serial Number : 1098

Number of samples tested : One

Date(s) of Test (s) : 28-10-2025 to 18-11-2025

CPRI Sample Code Number (s) : HVD25S0847

Particulars of tests conducted : Type Tests, All Routine Tests & Special Test (Refer Sheet 3 of 11)

Test in accordance with Standard / specification : IS 1180 (latest: yes)

Sampling plan : Not applicable

Customer's requirement : Nil

Deviations if any : Nil

Name of the witnessing persons

Customer's representative : Mr. Vivekanand Krishna (customer representative)

Other than customer's representative : None

Test subcontracted with address of the laboratory : None

Documents constituting this Report (In words)

Number of Sheet(s) : Eleven

Number of Oscillogram (s) : Ten

Number of Graph (s) : Two

Number of Photograph(s) : Nil

Number of Test Circuit Diagram(s) : One

Number of Drawing(s) : Two

(P. Naveenkumar)
Test Engineer

Reviewed and Authorized by



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DESCRIPTION OF SAMPLE TESTED

(As assigned by the manufacturer)

Test sample	:	3 Phase Distribution Transformer
Type	:	ONAN outdoor type distribution transformer
Serial number	:	1098
Rated power	:	250 kVA
Rated voltage - HV / LV	:	11000 / 433 V
Highest voltage of the equipment - Um	:	12 kV
Insulation levels		
HV LI / AC	:	LI 75 kV / AC 28 kV rms
LV LI / AC	:	AC 3 kV rms
Impedance (Percent)	:	4.5 %
Number of phases	:	3
Rated frequency	:	50 Hz
Vector group	:	Dyn11
Tappings	:	+5% to -10% in steps of 2.5% (7 taps)
Energy efficiency	:	NA
Type of cooling	:	ONAN
Maximum total loss at 50% load	:	980 W
Maximum total loss at 100% load	:	2930 W
Construction	:	Oil immersed, non-sealed, outdoor; LV helical/spiral copper, HV crossover copper; circular coils; CRGO core; tank plain/corrugated
Month & Year of Manufacture	:	10/2025
Volume of oil	:	460 litres

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SUMMARY OF TESTS CONDUCTED

1. Tests conducted : Refer the below table
2. Rating for which tested : Refer Sheet 5 of 11 to Sheet 10 of 11
3. Schedule of tests

The clause numbers of the standard IS 1180 (latest: yes) pertain to the test(s) conducted are detailed in the following table.

Tests Conducted	Clause Number (s)	Sheet
Special Tests		
Short-circuit withstand test (Ability to withstand the Dynamic Effects of Short circuit and Thermal Ability to withstand Short Circuit)	21.4 b)	7 of 11 & 8 of 11
No load current at 112.5 percent voltage	21.4 c)	10 of 11
Type Tests		
Temperature-rise test	21.3 b)	9 of 11
Pressure test	21.3 d)	10 of 11
All Routine Tests		
Measurement of winding resistance	21.2 a)	5 of 11
Measurement of voltage ratio and check of phase displacement	21.2 b)	5 of 11
Measurement of short-circuit impedance and load loss at 50% and 100% load	21.2 c)	5 of 11
Measurement of no-load loss and current	21.2 d)	5 of 11
Measurement of insulation resistance	21.2 e)	6 of 11
Induced over-voltage withstand test	21.2 f)	6 of 11
Separate source voltage withstand test	21.2 g)	6 of 11
Pressure test	21.2 h)	8 of 11
Oil leakage test	21.2 j)	8 of 11

4. Oscillogram number(s) : CPRI BRSCL25T1654S002, CPRI BRSCL25T1654S003, CPRI BRSCL25T1654S004, CPRI BRSCL25T1654S005, CPRI BRSCL25T1654S006, CPRI BRSCL25T1654S007, CPRI BRSCL25T1654S008, CPRI BRSCL25T1654S009, CPRI BRSCL25T1654S010 & CPRI BRSCL25T1654S011
5. Graph number(s) : CPRI BRSCL25T1654G01 & CPRI BRSCL25T1654G02
6. Photograph number(s) : Nil
7. Test circuit diagram number(s) : CRTL/SC/TR-07B
8. Drawing Numbers : Refer Sheet 4 of 11

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LIST OF DRAWINGS

Drawing Numbers

Manufacturer has guaranteed that the sample submitted for the test (s) has been manufactured in accordance with the following drawings.

Sl. No.	Drawing Number	Sheet Number	Revision Number
1	APT/250-11/001	-	-
2	APT/250-11/001B	-	-

It is verified that these drawings adequately represent the sample tested. Verification of this drawing by CPRI is limited to dimensional check only wherever possible.

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TEST RESULTS

ROUTINE TESTS

Measurement of winding resistance						
Tap switch position	Before SCW test			After SCW test		
	Principal (3)	Highest (1)	Lowest (7)	Principal (3)	Highest (1)	Lowest (7)
HV winding resistance at 75 °C (Ω) - average	7.101	7.504	6.346	7.098	7.502	6.343
LV winding resistance at 75 °C (m Ω) - average	2.317			2.318		

Measurement of voltage ratio							
Tap switch position	Tapping %	Before SCW test			After SCW test		
		U	V	W	U	V	W
1	+5	46.38	46.36	46.26	46.37	46.37	46.26
2	+2.5	45.22	45.22	45.12	45.23	45.23	45.13
3	0	44.1	44.09	44.06	44.11	44.09	44.07
4	-2.5	43.02	43.01	42.96	43.03	43.05	43.0
5	-5	41.92	41.92	41.92	41.93	41.92	41.92
6	-7.5	40.82	40.86	40.86	40.81	40.86	40.81
7	-10	39.73	39.72	39.72	39.73	39.72	39.72

Check of phase displacement	
Vector group was found to be Dyn11	

Measurement of short circuit impedance and load loss at 50% and 100% load						
Tap switch position	Before SCW test			After SCW test		
	Principal (3)	Highest (1)	Lowest (7)	Principal (3)	Highest (1)	Lowest (7)
% Impedance at 75 °C	4.38	4.442	4.487	4.387	4.446	4.496
Load loss at 75 °C (W)	2102.74	2056.03	2240.03	2101.21	2054.67	2239.94
Load loss at 50% of full load at 75 °C (W)	524.91	---	---	523.61	---	---

Measurement of no-load loss and current

Condition of the sample	Voltage (V)	Frequency (Hz)	No-load current (A)	No-load loss (W)
Before SCW test	433.06	49.95	1.689	383.18
After SCW test	433.06	49.99	1.687	383.26

SCW - Short-Circuit Withstand

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TEST RESULTS

ROUTINE TESTS (Contd...)

Check of Energy Efficiency Level

	Before SCW test	After SCW test
Total loss at 50% load (W)	908.09	906.87
Total loss at 100% load (W)	2485.92	2484.47

Measurement of insulation resistance

Insulation resistance of	Before SCW test (GΩ)	After SCW test (GΩ)
HV winding with respect to LV winding & tank connected together and earthed.	4.19	6.11
LV winding with respect to HV winding & tank connected together and earthed.	1.22	1.35
HV winding with respect to LV winding with tank earthed.	6.56	7.81

Induced over-voltage withstand test

Test on	Before SCW test	After SCW test
LV winding by keeping HV winding open circuit	866 V, 100 Hz, 60 s; current 1.58 A No disruptive discharge, withstood	866 V, 100 Hz, 60 s; current 1.66 A No disruptive discharge, withstood

Separate source voltage withstand test

Test on	Result (as logged)
a) HV winding and all terminals of LV winding & tank connected together to earth	28 kV for 60 s: No disruptive discharge, withstood
b) LV winding and all terminals of HV winding & tank connected together to earth	3 kV for 60 s: No disruptive discharge, withstood

SCW - Short-Circuit Withstand

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TEST RESULTS

SPECIAL TEST

SHORT-CIRCUIT WITHSTAND TEST

Test conditions

Source Short-circuit generator
Phases Three
Frequency 50 Hz

Test sample

Condition before test Transformer as after routine test
Mounting Transformer tank isolated and grounded through the protection current transformer

Connection

LV terminals Shorted
HV terminals Connected to source

Test details

Test circuit drawing number CRTL/SC/TR-07B

Short-circuit

Type Pre-set
Applied on LV terminals through non-inductive current measuring shunt
Short-circuit point Grounded

Short-circuit current calculation

Tap switch position	X/R at 75 °C	Required short-circuit current on LV side (kA)	
		rms	peak
Principal (3)	5.112	7.61	16.73
Highest (1)	5.31	7.5	16.61
Lowest (7)	4.907	7.43	16.2

Ability to Withstand the Dynamic Effects of Short-Circuit [Sub-clause 4.2 of IS: 2026 (Part-5):2011]

Oscillogram number	Current peak (kA)	Current rms (kA) (Average)	Duration (s)	Observation
Tap switch position no. Principal (3)				
CPRI BRSCL25T1654S002	16.8	7.53	0.54	No abnormality
CPRI BRSCL25T1654S003	16.82	7.66	0.54	No abnormality
CPRI BRSCL25T1654S004	16.74	7.71	0.54	No abnormality
Tap switch position no. Highest (1)				
CPRI BRSCL25T1654S005	16.89	7.52	0.54	No abnormality
CPRI BRSCL25T1654S006	16.78	7.54	0.54	No abnormality
CPRI BRSCL25T1654S007	16.73	7.53	0.54	No abnormality
Tap switch position no. Lowest (7)				
CPRI BRSCL25T1654S008	16.49	7.63	0.54	No abnormality
CPRI BRSCL25T1654S009	16.34	7.44	0.54	No abnormality
CPRI BRSCL25T1654S010	16.26	7.45	0.54	No abnormality
Thermal Ability to Withstand Short-Circuit [Sub-clause 4.1 of IS: 2026 (Part-5):2011] Tap switch position no. Principal (3)				
CPRI BRSCL25T1654S011	---	7.55	2.01	No abnormality

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Reactance values after transformer attains ambient temperature			
Tap switch position	% X before SCW test	% X after SCW test	% change in X
Principal (3)	4.299	4.306	-0.163
Highest (1)	4.365	4.369	-0.092
Lowest (7)	4.397	4.406	-0.205
Variation of reactance is within limits			
Negative value of % change in reactance indicates increase in the value & vice versa			

SCW - Short-Circuit Withstand

Physical Inspection:

Conductor, core & clampings: no visible damage

Spacers/support blocks: intact

Oil: clear

ROUTINE TESTS

OIL LEAKAGE TEST

Test Procedure : Transformer with all fittings including bushings in position, Air pressure was applied through breather connecting pipe.

Atmospheric pressure : NA

Ambient temperature : 25 °C

Head pressure calculated at base of the tank kPa (g)	Test Pressure at base of tank kPa (g)	Test duration (Hours)	Observations
14	28	8	No leakage at any point

Pressure (ROUTINE Test)

Test Procedure : Transformer tank fixed with all fittings including bushings in position, Air pressure applied through breather connecting pipe.

Atmospheric Pressure : NA

Ambient temperature : 25 °C

Air pressure above atmospheric pressure (kPa)	Duration (Minutes)	Observations
35	10	No leakage at any point

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TEST RESULTS

TYPE TEST

Temperature-rise test

Test method : Short-circuit

Test conditions

Source

Phases : Three

Frequency (Hz) : 50

Sample

HV terminals : Connected to three phase source

LV terminals : Shorted through copper links of negligible resistance

Tap position : Lowest Tap

Measured Losses

No load loss at rated voltage and at rated frequency (W)	383.26
Load loss at rated current at 75 °C at lowest tap (W)	2239.94

Test Procedure:

Step 1 - Total loss (2623.2 W) is injected until a steady state of oil temperature rise is established

Step 2 - The rated current (14.58 A) is injected for one hour, immediately after step-1.

The values of hot resistance of HV and LV windings at the time of shutdown are given in graphs

CPRI BRSCL25T1654G01 & CPRI BRSCL25T1654G02 respectively.

Ambient temperature: 22.85 °C

Measured Points	Temperature-rise (°C)	Guaranteed Limit declared by the manufacturer (°C)	Limit as per the standard (°C)	Observations
Top oil	26.01 (as logged)	35	35	Within limits
HV winding	39.7 (calculated)	40	40	Inconclusive (close to the limit)
LV winding	34.3 (calculated)	40	40	Within limits

Winding temperature rise not logged; calculated from the logged hot and cold resistances.

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Pressure (TYPE TEST)

Test Procedure : Transformer tank fixed with all fittings including bushings in position. First air and then vacuum applied through breather connecting pipe.

Atmospheric Pressure : NA

Ambient temperature : 25 °C

Air Pressure test:

Air pressure above atmospheric pressure (kPa)	Duration (Minutes)	Permanent deflection of flat plate after release of air pressure (mm)	Observations
80	30	1.08	Within limits

Vacuum test:

Vacuum in level of mm of Mercury	Duration (Minutes)	Permanent deflection of flat plate after release of vacuum (mm)	Observations
500	30	2.87	Within limits

SPECIAL TESTS

No load current at 112.5 percent voltage

Test Procedure	No-load current of transformer was measured at rated voltage and 112.5 % rated voltage at 50 Hz on LV side. Values are given below.		
Average no-load current (Percentage of rated full load current)			
At 100 percent rated voltage (433.06 V)		At 112.5 percent rated voltage (487.32 V)	
Permissible value given in the standard	Measured value	Permissible value given in the standard	Measured value
2	0.51	5	1.28

Conclusion: The sample tested complies with the requirement(s) of Clause(s) referred of IS 1180 (latest: yes) for the tests conducted.

* Items flagged by automated validation - to be confirmed by the reviewer before approval:

Check	Detail
Identifier consistency	Losses logsheet (resistance): transcribed '25T1656' / 'HVD25S0847' but work instruction = 25T1654 / 0847. Verify handwriting (4/6/H).
Identifier consistency	Losses logsheet (no-load): transcribed '25T1656' / 'HVD25S0847' but work instruction = 25T1654 / 0847. Verify handwriting (4/6/H).
Identifier consistency	Routine test logsheet: transcribed '25T1654' / 'HVD25S0867' but work instruction = 25T1654 / 0847. Verify handwriting (4/6/H).
HV winding temperature rise	39.7 K calculated (limit 40 K, margin 0.3 K: inconclusive)
Correction factor	Written as 0.275 at top of page 2 but 0.475 used in the formula - confirm

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NOTE

1. The Test results relate only to the sample(s) tested.
2. Publication or reproduction of this Test Report in any form other than by complete set of the whole Test Report and in the language written is not permitted without the written consent of CPRI.
3. Any Corrections / erasure invalidate the Test Report.
4. Any anomaly / discrepancy in the Test Report should be brought to notice of CPRI within 45 days from the date of issue.
5. All documents constituting the Test Report are stitched together with a continuous silk thread, the two ends of which have been brought over the front sheet of the Test Report and sealed with a CPRI logo printed paper sticker
6. NABL has Accredited this laboratory as per ISO/IEC 17025-2017 vide Report no.TC-5452 for the tests carried out.



TC-5452

-----End of Test Report-----

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